

Syllabus
MWI10015 Management by Key Performance Indicators
Prof. Dr. Bingwen Yan
Summer Semester 2026

Level	Master	
Credits	3	
Student Contact Hours	2	
Workload	90 hours	
Prerequisites	Bachelor level knowledge in financial accounting and control and value based management	
Language of teaching	English	
Time	See online schedule LSF	
Room	See online schedule LSF	
Start Date	Thursday, 09 April 2026	
Lecturer(s)	Name	Prof. Bingwen Yan
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	Office Hours	See online schedule LSF
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Summary

This course provides advanced knowledge in financial accounting and financial control, integrating essential tools in treasury management, cash flow analysis, corporate taxation, and investor relations. Delivered through a blend of lectures, interactive workshops, case studies, and student presentations, it equips learners with both the theoretical understanding and the practical application required for modern financial leadership in a technology-driven environment. A central focus of the course is the development of skills in designing, analysing, and communicating Key Performance Indicator (KPI) systems for managerial decision-making. Key areas of emphasis include understanding KPI frameworks and performance logic, and using financial statements as primary sources for KPI development. Students will learn to derive and interpret cash flow and value-based performance metrics that directly link to corporate value creation. The curriculum also explores the role of technology in modern finance, introducing students to digital and automated controlling systems that enable real-time data analysis and reporting. Furthermore, the course addresses the critical function of performance communication through investor and stakeholder reporting, and examines the unique challenges of crisis performance management and organisational resilience in volatile economic environments.

This course represents the first part of the "Value Based Management" module. It establishes the foundational KPI knowledge and performance measurement skills necessary for effective management control. Building upon this foundation, the second part of the module, "Financial Management & Control", will apply these concepts within a business simulation and through student presentations on advanced controlling topics.

Key areas of emphasis include:

- KPI frameworks and performance logic
- Financial statements as sources for KPI development
- Cash flow and value-based performance metrics
- Digital and automated controlling systems
- Investor and stakeholder reporting
- Crisis performance management and organisational resilience

Outline of the Course

The course has a workshop character with lectures, students' presentations, case studies, discussions in class, and presentations from outside experts. A solid knowledge of finance, financial control, and capital markets is a prerequisite. Depending on the knowledge and the interests of the students, discussions can develop around the following course topics:

1. **Foundations of KPI Management (Workshops 1–2):** We begin by establishing the strategic role of KPIs in value-based management. Students will explore the distinction between financial and managerial accounting, and analyse the balance sheet as a primary instrument for deriving core financial KPIs (liquidity, leverage, working capital).
2. **Financial Statements as KPI Sources (Workshops 3–5):** This block focuses on extracting and interpreting performance metrics from the profit & loss statement (gross margin, EBIT, ROA) and the cash flow statement (free cash flow, cash conversion cycle). Students will learn to connect financial data to solvency, liquidity, and overall financial resilience.
3. **KPI Frameworks and Reporting Standards (Workshops 6–8):** We then turn to the design of integrated KPI systems. Topics include the impact of different reporting standards (IFRS, US-GAAP) on KPI comparability, and the role of digital tools (ERP, BI systems) in automating reporting processes and enabling real-time decision-making.
4. **Value-Based and Crisis Performance Management (Workshops 9–12):** The course concludes with advanced topics in value-based management, including Economic Value Added (EVA) and return figures (ROI, ROCE, ROE). We will also examine the role of KPI systems during financial crises (2008/09, Euro Crisis 2011) and discuss how performance management contributes to organisational resilience.

Course Intended Learning Outcomes and their Contribution to Program Intended Learning Outcomes / Program Goals

Program Intended Learning Outcomes <i>After completion of the program the students will be able...</i>	Course Intended Learning Outcomes <i>After completion of the course the students will be able...</i>
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1 Responsible leadership in organisational contexts

1.1 ...to demonstrate their solid knowledge of numerous relevant management principles. They are able to explain and discuss them discerningly.	...to adequately use key figures to control a company and to explain their relevance discerningly.
1.2 ...to apply management principles within an organisational context.	...to apply concepts of global business analysis and management to real-world company situations.

2 Creative problem-solving skills in a complex business environment

2.1 ...to recognise and define problems as well as assess their importance.	...to lead technical discussions on topics relevant to controlling and assess problem significance.
2.2 ...to analyse complex in-company and inter-company problems and challenges from different perspectives and/or within an international context.	...to use company key figures for target-oriented control and to know their benefits and limitations.
2.3 ...to clarify successfully complex problems and solutions to both experts and laymen.	...to prepare and deliver presentations and scientific seminar papers on controlling topics.

3 Creative problem-solving skills in a complex business environment

3.1 ...to successfully apply research methods relevant to engineering and management.	...to write seminar papers in compliance with the principles of scientific work.
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4 Interdisciplinary and integrative work

4.1 ...to apply their expert knowledge in the engineering and business field and to provide integrated solutions to complex tasks.	...to use controlling knowledge to evaluate technical concepts regarding their economic sense.
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Academic Integrity and Student Responsibility

Students are encouraged to form study groups and discuss the course material, including the topics for presentations and papers. However, the submitted seminar paper must be your own original work. Discussions should aim to deepen understanding, not to circumvent the individual learning process. All sources used must be properly cited. Failure to do so constitutes academic dishonesty.

Teaching Philosophy

My primary goal is to help you develop a practical and critical understanding of how KPIs are used to manage and lead a business. I believe in active learning and real-world application. I encourage you to bring your own questions and experiences from your professional background into the classroom. If you encounter any difficulties with the material, your presentation, or the seminar paper, please contact me as early as possible. I am here to support your learning journey and am committed to your success. The earlier you reach out, the more effectively I can assist you.

Literature and Course Materials

- Annual reports of leading international companies (e.g. SAP, Bosch, Daimler).
- Atrill, P., McLaney E. (2025), Accounting and Finance for Non-Specialists, 13th ed., Harlow: Pearson.
- Buchholz R. (2016), Grundzüge des Jahresabschlusses nach HGB und IFRS, 9. Auflage, München: Vahlen.
- Robert, C. Higgins, L. Koski Jennifer, and Mitton Todd. (2023). Analysis for financial management. McGraw-Hill Education.
- Horváth, P., Gleich, R., Seiter, M. (2020), Controlling. 14. Aufl., München: Vahlen.
- Horváth & Partner (2017, Hrsg.), Das Controllingkonzept, 9. Aufl., München: Dtv.
- Kieso, D.E., Weygandt, J.J., Warfield, T.D. (2022), Intermediate Accounting, 18th edition, New York: Wiley. ISBN: 978-1-119-77889-9
- Weber, J., Schäffer U. (2016), Einführung in das Controlling, 15. Aufl., Stuttgart: Schäffer Poeschel.
- Weygandt, J.J., Paul D. Kimmel, P. D., Mitchell, J. E., (2023). Financial Accounting, 12th edition, New York: Wiley.
- Weygandt, J.J., Kieso, D.E., Kimmel, P.D. (2020), Managerial Accounting: Tools for Business Decision-Making, 6th edition, New York: Wiley.
- Hillier, D. (2024). Corporate Finance 5th edition. McGraw Hill eBook. 1526849917 · 9781526849915
- Ross, S., Westerfield, R.W, Jaffe, J., Jordan, D., & Shue, K. (2025 Release ISE). Corporate Finance. McGraw-Hill. ISBN:1266961003 · 9781266961007.
- Correia, C. (2023). Financial Management. 10th edition. Juta. ISBN: 9781998963133.
- EY/ PwC annual reports on digital controlling.